Jaafar Alhourani

Email : jalhourani3@gmail.com | **Phone :** +963 0980 329 517 | **Address :** Homs, Syria

Website : **https://jaafar-a-l-hourani.onrender.com** | **GitHub:** https://github.com/AL-Hourani

Personal Profile

Motivated and self-driven student seeking to pursue a Master's degree in Artificial Intelligence and Machine Learning. During my academic studies, I developed strong self-learning abilities and built a solid technical background. I initially focused on web development and completed several projects in this field. Later, I shifted my focus toward Machine Learning, Artificial Intelligence, and Computer Vision, aiming to integrate these technologies to build intelligent applications and smart web platforms.

Education

University of Homs: BSc Computer Science **Sep 2020 – Sep 2025**

Major / Specialization: Software Engineering & Information Systems

GPA : 3.1 / 4.0 (Equivalent to 75.96%)

Relevant Modules : Software Engineering , Database Systems , Data Structures & Algorithms , Machine Learning , Probability and Statistics

Research Experience

Public Transport Monitoring and Tracking System **Sep 2023**

- Designed and developed a real-time GPS-based monitoring system for public transport
- Implemented data collection, processing
- Visualization on maps to track vehicles and detect deviations from planned roads

Relevant Research Skills

- **Programming Languages:** Python, JavaScript, Go – proficient in coding, debugging, and developing efficient solutions.
- **Data Analysis & Visualization:** Skilled in Python data analysis libraries (Pandas, NumPy, Matplotlib) for processing and interpreting datasets.
- **Machine Learning & AI:** Knowledgeable in machine learning using Scikit-Learn; capable of building predictive models and analyzing patterns.
- **Web & Interactive UI Development:** Experienced in building interactive interfaces using React framework.
- **Database Design & Management:** Proficient in designing, implementing, and managing relational databases.
- **Teamwork & Collaboration:** Strong ability to work effectively in research and development teams, contributing to problem-solving and project delivery.
- **Research & Problem-Solving:** Competent in designing experiments, analyzing real-time data, and presenting research findings clearly

Academic Projects

Diabetes Management System (Graduation Project) Sep – 2025

Developed a comprehensive system to assist diabetes patients in monitoring their health and improving their care. The system includes three versions:

- **Healthcare Centers :** Allows staff to register patients, track health metrics, and log data over time during visits
- **Supervisor :** Monitors medication inventory across centers and responds to inquiries

- **Patient :** Enables patients to track their own health data and receive guidance
- **Technologies Used :** React (UI/UX design), Progressive Web App (PWA) for cross-platform access, Go (API development), PostgreSQL (database management) , Machine Learning

Machine Learning & Medical AI Projects 2025 – Present

Designed and implemented multiple machine learning and deep learning solutions across real-world and research-oriented applications, with a growing focus on medical image analysis and explainable AI systems.

• Brain Tumor Detection (CNN-Based)

Developed a deep learning model to classify brain MRI images into tumor and non-tumor categories using Convolutional Neural Networks (CNNs). Applied image preprocessing, feature extraction, and model optimization techniques to improve classification performance.

• Medical Anomaly Detection & Patient Risk Monitoring

Built an intelligent system for detecting abnormal patterns in clinical data using unsupervised learning techniques. Implemented real-time risk monitoring and interactive 3D visualizations using Streamlit.

• Explainable AI Dashboard

Designed an interactive dashboard to interpret machine learning predictions using SHAP and LIME, enabling transparent model analysis and feature importance visualization.

- **Explainable AI for Chronic Kidney Disease (CKD)**

Developed a full-stack machine learning system for predicting chronic kidney disease using clinical datasets. Integrated explainability tools (SHAP) and deployed results through an interactive Streamlit interface.

- **Core Machine Learning Projects**

Implemented foundational models including:

- Handwritten Digit Recognition (classification)

- House Price Prediction (regression)

- Classification projects

- Various classification and clustering tasks across multiple datasets

Technologies: Python, Scikit-Learn, TensorFlow/Keras, Pandas, NumPy, Matplotlib, Streamlit, SHAP,

Languages

Arabic: Native **English: Very Good (Academic)**

References : Available upon request